

Microorganisms are measured in micrometers and nanometers. The average bacterial cell is 0.001mm in diameter.

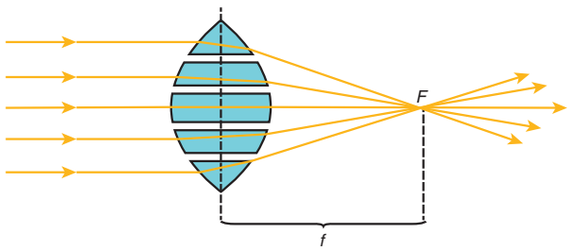


Figure 2.4: Lens function

Generating an image with a lens

When an object is placed outside the focal plane (the plane containing the focal point of the lens), all the light rays from the object are bent by the lens. The bent rays converge at the opposite focal point. At the focal point, the light rays continue and converge with nonparallel refracted light rays. The resultant reversed and magnified image is formed in the plane of convergence (Figure 2.5).

Microscope resolution

Objective is the important part in the microscope which is responsible to produce a clear image. The resolution of the objective is most important. Resolution is the capacity of a lens to separate or distinguish between small objects that are close together. The major factor in the resolution is the wave length of light used. The greatest resolution obtained with light of the shortest wave length, that is the light at the blue end of the visible spectrum are in the range of 450 to 500nm. The highest resolution possible in compound light microscope is about 0.2µm. That means, the two objects closer together than 0.2µm are not resolvable as distinct and separate. The light microscope is equipped with three or four objectives. The working distance of an objective is the distance between the front surface of the lens and the surface of the cover glass or the specimen. Objectives with large numerical apertures and great resolving power have short working distances.

Numerical aperture

Numerical Aperture (NA) is the value representing the light gathering capacity of an objective lens. NA was first described

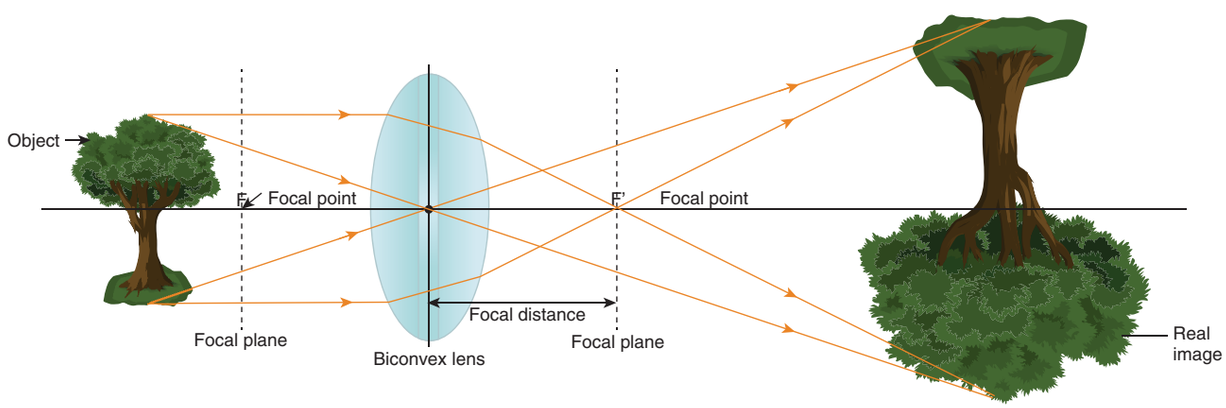


Figure 2.5: Generating an image with a lens

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